



Data Structure

Linear: Elements form a sequence, i.e., liner list

Array: Elements in sequential memory locations

Link list: Relationship between elements by means of pointer.

Nonlinear: Elements do not form a sequence

Tree

Graph

These two will be defined later.



Data Structure

Linear: Elements form a sequence, i.e., liner list

Array: Elements in sequential memory locations

Link list: Relationship between elements by means of pointer.

- (a) *Traversal.* Processing each element in the list.
- (b) *Search.* Finding the location of the element with a given value or the record with a given key.
- (c) *Insertion.* Adding a new element to the list.
- (d) *Deletion.* Removing an element from the list.
- (e) *Sorting.* Arranging the elements in some type of order.
- (f) *Merging.* Combining two lists into a single list.

Operations that are performed on liner data structure.



Data Structure

Linear: Elements form a sequence, i.e., liner list

Array: Elements in sequential memory locations

Link list: Relationship between elements by means of pointer.

Our target is to explain it first.

Nonlinear: Elements do not form a sequence

Tree

Graph

These two will be defined later.



Data Structure

Linear: Elements form a sequence, i.e., liner list

Array: Elements in sequential memory locations

Link list: Relationship between elements by means of pointer.

Liner Array

Multi-dimensional Array

Nonlinear: Elements do not form a sequence

Tree

Graph

These two will be defined later.



CSE 203: Data Structure

Data Structure

Linear Array: This one dimensional array.

List of finite n homogenous data elements.

$A[1], A[2], A[3], \dots, A[N]$

$A_1, A_2, A_3, \dots, A_n$

$A(1), A(2), \dots, A(N)$

Representation style



CSE 203: Data Structure

Data Structure

Linear Array: This one dimensional array.

List of finite n homogenous data elements.

$A[1], A[2], A[3], \dots, A[N]$

Elements	15	12	0	11	20	11	9	28	33
Index	1	2	3	4	5	6	7	8	9

Elements	15	12	0	A	20	11	9	D	33
Index	1	2	3	4	5	6	7	8	9



CSE 203: Data Structure

Data Structure

Linear Array: This one dimensional array.

List of finite n homogenous data elements.

$A[1], A[2], A[3], \dots, A[N]$

Elements	15	12	0	11	20	11	9	28	33
Index	1	2	3	4	5	6	7	8	9

Elements	15	12	0	A	20	11	9	D	33
Index	1	2	3	4	5	6	7	8	9



Not array



CSE 203: Data Structure

Data Structure

Linear Array: This one dimensional array.

List of finite n homogenous data elements.

Elements	15	12	0	11	20	11	9	28	33
Index	1	2	3	4	5	6	7	8	9

Lower bound (LB)

: Smallest index

Upper bound (UB)

: Largest index

$$\text{Length} = \text{UB} - \text{LB} + 1$$



CSE 203: Data Structure

Data Structure

Linear Array: This one dimensional array.

Let DATA be a 6-element linear array of integers such that
DATA[1] = 247 DATA[2] = 56 DATA[3] = 429 DATA[4] = 135 DATA[5] = 87
DATA[6] = 156

Sometimes we will denote such an array by simply writing
DATA: 247, 56, 429, 135, 87, 156

DATA	
1	247
2	56
3	429
4	135
5	87
6	156

(a)

DATA					
247	56	429	135	87	156
1	2	3	4	5	6

(b)



CSE 203: Data Structure

Data Structure

Linear Array: This one dimensional array.

An automobile company uses an array AUTO to record the number of automobiles sold each year from 1932 through 1984. Rather than beginning the index set with 1, it is more useful to begin the index set with 1932 so that

$AUTO[K]$ = number of automobiles sold in the year K

Then $LB = 1932$ is the lower bound and $UB = 1984$ is the upper bound of AUTO. By Eq. (4.1),

$$\text{Length} = UB - LB + 1 = 1984 - 1932 + 1 = 53$$

That is, AUTO contains 53 elements and its index set consists of all integers from 1932 through 1984.

Sold Auto	15	12	50	11	20	11	19	28	33
Year	1932	1933	...	...	...	...	...	1983	1984

Lower bound (LB)

Upper bound (UB)

$$\text{Length} = UB - LB + 1 = 1984 - 1932 + 1 = 53$$



Linear Array: Representation in memory

	LA
1000	
1001	
1002	
1003	

- Memory of the computer is simply sequence of address.
- Hence, linear array holds a block of contiguous memory cells.
- Let LA is a linear array
 - Start at memory location 1000, that is, first data at 1000
 - Second data at 1001
 - Third data at 1002
 - and so on.

$LA[K]$ = Element in the K^{th} location of the array LA.

Here, $K \in \{1000, 1001, 1002, 1003, \dots, \dots\}$

$LOC(LA[K])$ = Address of the element $LA[K]$ in the array LA.

Computer keeps track only of the first element of the array LA

Address of the first element of the array LA = $Base(LA)$



CSE 203: Data Structure

Data Structure

Linear Array: Representation in memory

	LA
1000	
1001	
1002	
1003	

Address of the first element of the array LA = $Base(LA)$

$Loc(LA[K]) = Base(LA) + K - LB$

$$LOC(LA[K]) = Base(LA) + w(K - \text{lower bound})$$

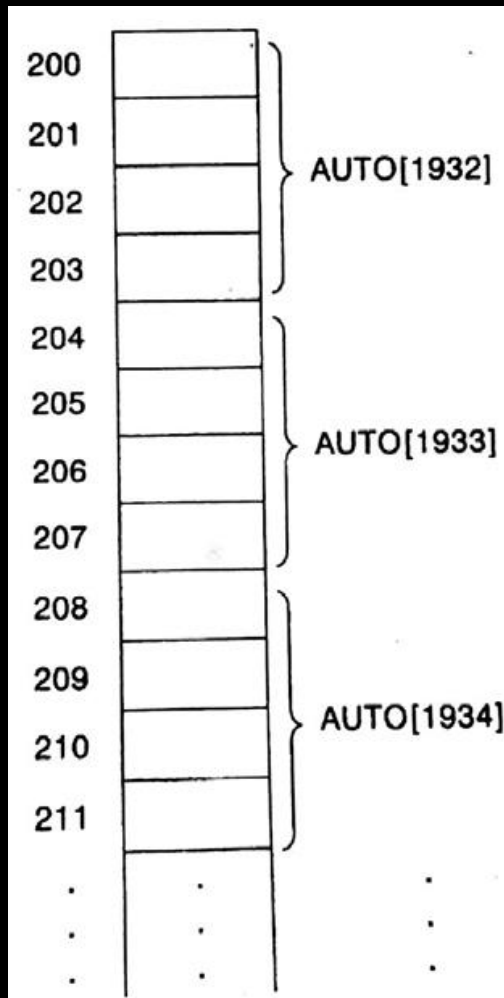
Where w is the number of words per memory cell.



CSE 203: Data Structure

Data Structure

Linear Array: Representation in memory



Address of the first element of the array $LA = \text{Base}(LA)$

$$\text{LOC}(LA[K]) = \text{Base}(LA) + w(K - \text{lower bound})$$

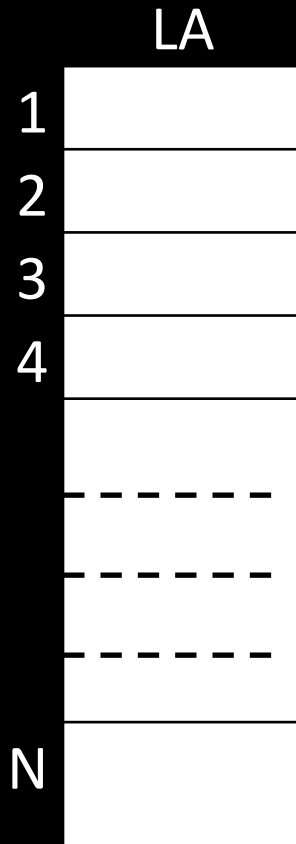
Here $w=4$



Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

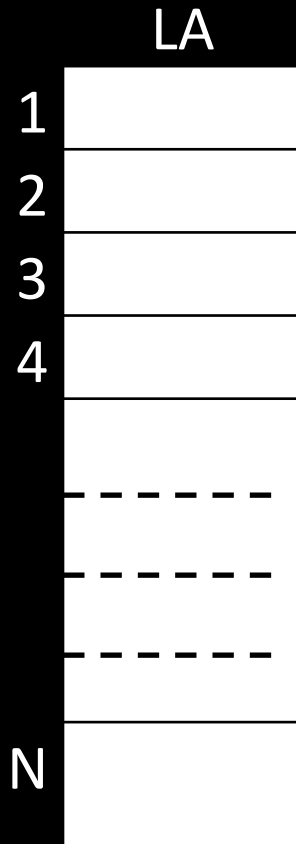
1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$.
3. [Visit element.] Apply PROCESS to $LA[K]$.
4. [Increase counter.] Set $K := K + 1$.
[End of Step 2 loop.]
5. Exit.



Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



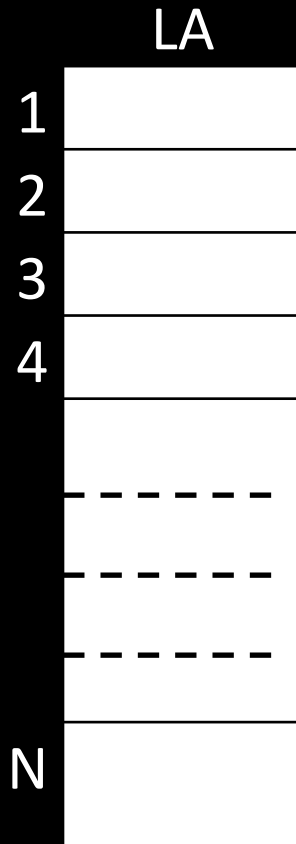
Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$. ←
 2. Repeat Steps 3 and 4 while $K \leq UB$.
 3. [Visit element.] Apply PROCESS to $LA[K]$.
 4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=LB=1$



Linear Array: Traversing



Accessing and processing of each element of array LA

- exactly once and
- sequentially

Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
 2. Repeat Steps 3 and 4 while $K \leq UB$. ←
 3. [Visit element.] Apply PROCESS to $LA[K]$.
 4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=1$

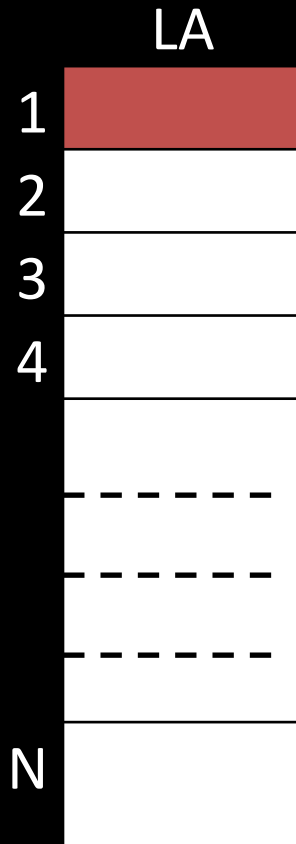
If $N=10$, $UB=10$
Then $K \leq UB$ means $1 \leq 10$ and
thus, it is true



Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
 2. Repeat Steps 3 and 4 while $K \leq UB$.
 3. [Visit element.] Apply PROCESS to $LA[K]$.
 4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=1$

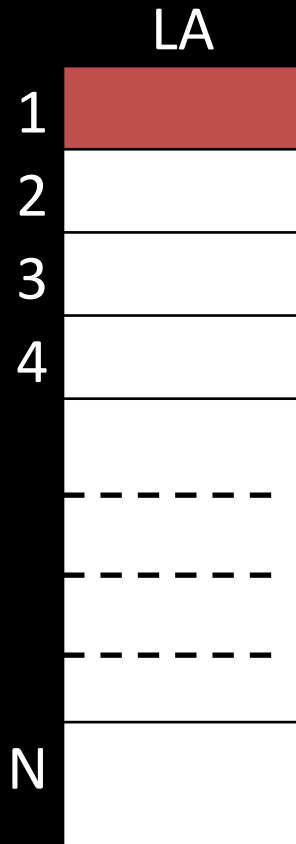
If $N=10$, $UB=10$
Then $K \leq UB$ means $1 \leq 10$ and
thus, it is true



CSE 203: Data Structure

Data
Structure

Linear Array: Traversing



Accessing and processing of each element of array LA

- exactly once and
- sequentially

Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$.
3. [Visit element.] Apply PROCESS to $LA[K]$.
4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=1$

$K=K+1$ means $K=2$

If $N=10$, $UB=10$

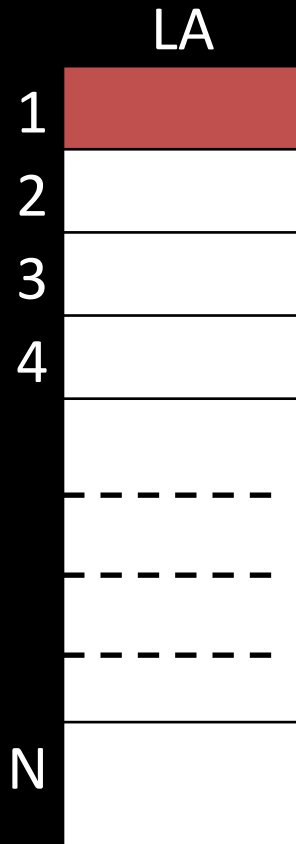
Then $K \leq UB$ means $1 \leq 10$ and thus, it is true



CSE 203: Data Structure

Data Structure

Linear Array: Traversing



Accessing and processing of each element of array LA

- exactly once and
- sequentially

Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
 2. Repeat Steps 3 and 4 while $K \leq UB$. ←
 3. [Visit element.] Apply PROCESS to $LA[K]$.
 4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=2$

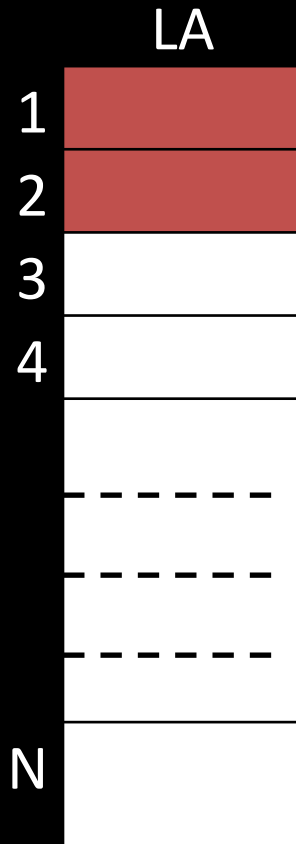
$K \leq UB$ means $2 \leq 10$ and thus, it is true



CSE 203: Data Structure

Data
Structure

Linear Array: Traversing



Accessing and processing of each element of array LA

- exactly once and
- sequentially

Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$.
3. [Visit element.] Apply PROCESS to LA[K]
4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=2$

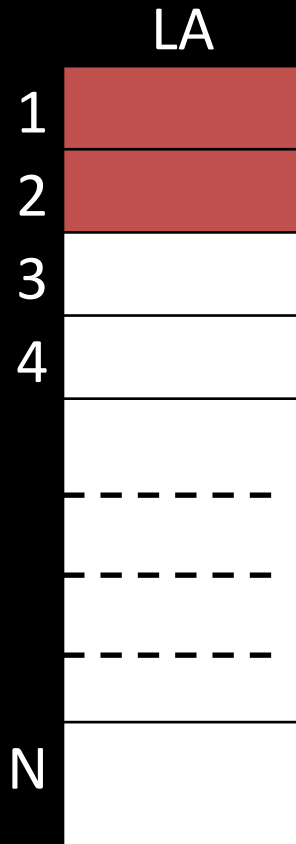
$K \leq UB$ means $2 \leq 10$ and thus, it is true



CSE 203: Data Structure

Data
Structure

Linear Array: Traversing



Accessing and processing of each element of array LA

- exactly once and
- sequentially

Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

```
1. [Initialize K := LB]
2. R ← LA[K]
3. PROCESS(R)
4. [Increase counter.] Set K := K + 1.
   [End of Step 2 loop.]
5. Exit.
```

This way it will continue and consider that the repetition is done for 9 times, that is $K=10$

$K=2$

$K=K+1$ means $K=3$

$K \leq UB$ means $2 \leq 10$ and thus, it is true



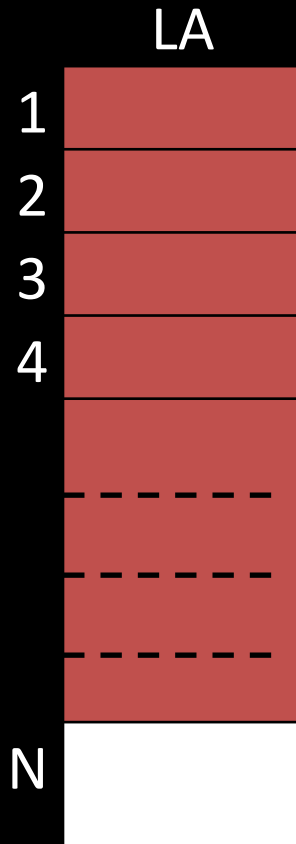
CSE 203: Data Structure

Data Structure

Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$. ←
3. [Visit element.] Apply PROCESS to $LA[K]$.
4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=10$

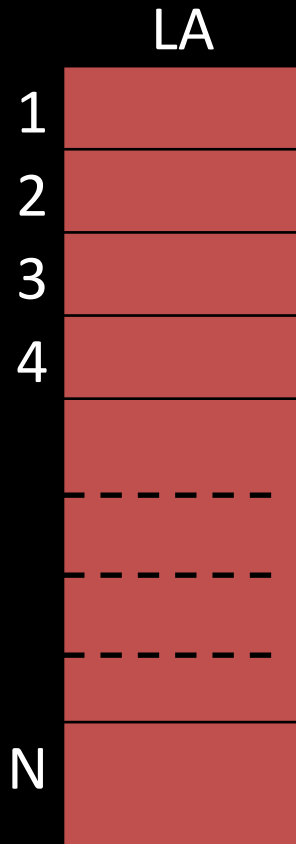
$10 \leq 10$ and thus, it is true



Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$.
3. [Visit element.] Apply PROCESS to $LA[K]$
4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=10$

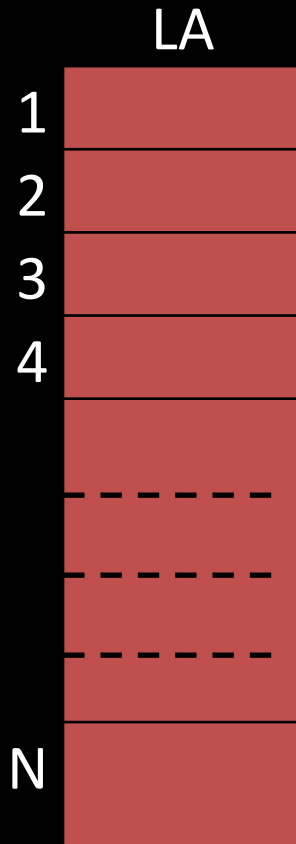
$10 \leq 10$ and thus, it is true



Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$.
3. [Visit element.] Apply PROCESS to $LA[K]$.
4. [Increase counter.] Set $K := K + 1$.
5. Exit.

$K=10$

$10 \leq 10$ and thus, it is true

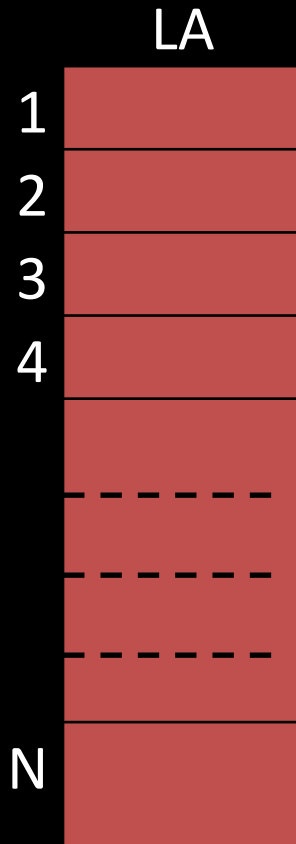
$K=K+1$ means $K=11$



Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$. ←
3. [Visit element.] Apply PROCESS to $LA[K]$.
4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

$K=11$

$11 \leq 10$ and thus, it is false



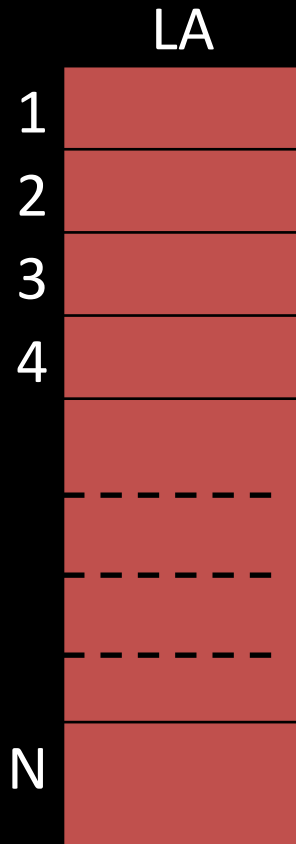
CSE 203: Data Structure

Data Structure

Linear Array: Traversing

Accessing and processing of each element of array LA

- exactly once and
- sequentially



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$.
3. [Visit element.] Apply PROCESS to $LA[K]$.
4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit. ←

$K=11$

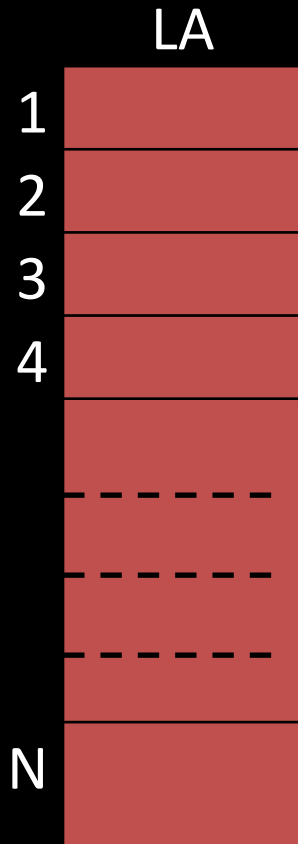
$11 \leq 10$ and thus, it is false



CSE 203: Data Structure

Data
Structure

Linear Array: Traversing



Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. [Initialize counter.] Set $K := LB$.
2. Repeat Steps 3 and 4 while $K \leq UB$.
3. [Visit element.] Apply PROCESS to $LA[K]$.
4. [Increase counter.] Set $K := K + 1$.
- [End of Step 2 loop.]
5. Exit.

Algorithm 1: Here LA is a linear array with lower bound LB upper bound UB. The algorithm traverses LA by applying PROCESS to each element of LA.

1. Repeat for $K = LB$ to UB :
 Apply PROCESS to $LA[K]$.
 [End of loop.]
2. Exit.

**Alternate
Algorithm**



CSE 203: Data Structure

Data
Structure

Linear Array: Traversing

	SL
Roll	CGPA
1	3.5
2	3.25
3	3.5
4	

	3.75

	3.9

N	3.00

Is it possible to use traversing algorithm (Algorithm Traverse) in counting the number of students in JKKNIU whose CGPA is greater than 3.5 in 2018?

Algorithm Traverse

1. Repeat for $K = LB$ to UB :
 Apply PROCESS to $LA[K]$.
 [End of loop.]
2. Exit.

Algorithm CGPA: Let SL of size N is an array of student list. Its index means the roll no of a student and the value of each index contains their CGPA till 2018. Use T to count the number of students whose $CGPA \geq 3.5$. Here, $LB=1$ and $UB=N$.

Step 1: $T=0$ [Initialization]

Step 2: Repeat for $K=LB$ to UB

Step 3: If $SL[K] \geq 3.5$ then $T=T+1$;
 [End of loop]

Step 4: Return;



CSE 203: Data Structure

Data
Structure

Linear Array: Traversing

SL

Roll CGPA

1 3.5

2 3.25

3 3.5

4

3.75

3.9

3.00

N

Let results of all the departments are stored in SL sequentially and the Dept's LB (DLB) and Dept's UB (DUB) of each dept is known. How to process for a specific dept?

Algorithm CGPA: Let *SL* of size *N* is an array of student list. Its index means the roll no of a student and the value of each index contains their CGPA till 2018. Use *T* to count the number of students whose $CGPA \geq 3.5$. Here, $LB=1$ and $UB=N$.

Step 1: $T=0$ [Initialization]

Step 2: Read the value of DLB and DUB

Step 3: Repeat for $K=DLB$ to DUB

Step 4: If $SL[K] \geq 3.5$ then $T=T+1$;
[End of loop]

Step 5: Return;

Algorithm Traverse

1. Repeat for $K = LB$ to UB :
 Apply PROCESS to $LA[K]$.
 [End of loop.]
2. Exit.



Linear Array: Insertion and Deletion

Scenario 1: Simply adding at the end of the array

Suppose TEST has been declared to be a 5-element array but data have been recorded only for TEST[1], TEST[2] and TEST[3]. If X is the value of the next test, then one simply assigns

$$\text{TEST}[4] := X$$

to add X to the list. Similarly, if Y is the value of the subsequent test, then we simply assign

$$\text{TEST}[5] := Y$$

to add Y to the list. Now, however, we cannot add any new test scores to the list.



CSE 203: Data Structure

Data Structure

Linear Array: Insertion and Deletion

Scenario 1: Simply adding at the end of the array

Suppose TEST has been declared to be a 5-element array but data have been recorded only for TEST[1], TEST[2] and TEST[3]. If X is the value of the next test, then one simply assigns

Scenario 2: Insertion and deletion in/from a sorted list.

NAME	
1	Brown
2	Davis
3	Johnson
4	Smith
5	Wagner
6	
7	
8	

(a)

NAME	
1	Brown
2	Davis
3	Ford
4	Johnson
5	Smith
6	Wagner
7	
8	

(b)

NAME	
1	Brown
2	Davis
3	Ford
4	Johnson
5	Smith
6	Taylor
7	Wagner
8	

(c)

NAME	
1	Brown
2	Ford
3	Johnson
4	Smith
5	Taylor
6	Wagner
7	
8	

(d)



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

. LA

1 10

2 12

3 11

4 15

5 21

6 22

7 19

8 20

N 14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = La[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

While executing

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

$N = 9$
 $J = 9$
For example $K = 7$



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$

$J = 9$

$K = 7$

$J \geq K$ means $9 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
J	14
J+1	

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] := LA[J]$ ← While executing

Step 4: Set $J := J - 1$

Step 5: Set $LA[K] := \text{ITEM}$

Step 6: Set $N := N + 1$

Step 7: Exit

$N = 9$

$J = 9$

$K = 7$

$J \geq K$ means $9 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
J	14
J+1	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] := LA[J]$ ← While executing

Step 4: Set $J := J - 1$

Step 5: Set $LA[K] := \text{ITEM}$

Step 6: Set $N := N + 1$

Step 7: Exit

$N = 9$

$J = 9$

$K = 7$

$J \geq K$ means $9 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
J	14
J+1	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$
 $J = 9 - 1 = 8$
 $K = 7$

$J \geq K$ means $9 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
J	20
J+1	14
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$
 $J = 9 - 1 = 8$
 $K = 7$

$J \geq K$ means $9 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
J	20
J+1	14
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$ ← While executing
//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

$N = 9$
 $J = 9 - 1 = 8$
 $K = 7$

$J \geq K$ means $8 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
J	20
J+1	14
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

While executing

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

$N = 9$
 $J = 9 - 1 = 8$
 $K = 7$

$J \geq K$ means $8 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
J	20
J+1	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$ ← While executing

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

$N = 9$
 $J = 9 - 1 = 8$
 $K = 7$

$J \geq K$ means $8 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
J	20
J+1	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$
 $J = 8 - 1 = 7$
 $K = 7$

$J \geq K$ means $8 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	19
J+1	20
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$
 $J = 8 - 1 = 7$
 $K = 7$

$J \geq K$ means $8 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	19
J+1	20
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$ ← While executing
//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

$N = 9$

$J = 7$

$K = 7$

$J \geq K$ means $7 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	19
J+1	20
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$ ← While executing

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

$N = 9$

$J = 7$

$K = 7$

$J \geq K$ means $7 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	19
J+1	19
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$ ← While executing

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

$N = 9$

$J = 7$

$K = 7$

$J \geq K$ means $7 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	19
J+1	19
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$
 $J = 7 - 1 = 6$
 $K = 7$

$J \geq K$ means $7 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
J	22
J+1	19
8	19
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$
 $J = 7 - 1 = 6$
 $K = 7$

$J \geq K$ means $7 \geq 7$ is true



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
J	22
J+1	19
8	19
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

$N = 9$

$J = 6$

$K = 7$

$J \geq K$ means $6 \geq 7$ is False



CSE 203: Data Structure

Data Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	19
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

ITEM=18

N=9

J=6

K=7

$J \geq K$ means $6 \geq 7$ is False



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	18
8	19
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

ITEM=18

N=9

J=6

K=7

$J \geq K$ means $6 \geq 7$ is False



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	18
8	19
N	20
	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$ ← While executing

Step 7: Exit

ITEM=18
 $N=9+1=10$
 $J=6$
 $K=7$

$J \geq K$ means $6 \geq 7$ is False



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	18
8	19
9	20
N	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to N^{th} elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$ ← While executing

Step 7: Exit

ITEM=18

N=10

J=6

K=7

$J \geq K$ means $6 \geq 7$ is False



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	18
8	19
9	20
N	14

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

//Move J to Nth elements to their next position and start it from the last

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = \text{ITEM}$

Step 6: Set $N = N + 1$

Step 7: Exit

While executing

ITEM=18

N=10

J=6

K=7

$J \geq K$ means $6 \geq 7$ is False



Linear Array: Insertion and Deletion

Algorithm INSERT:

Algorithm INSERT (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm inserts an element ITEM into the Kth position in LA.

Step 1: Set $J := N$

Step 2: Repeat Steps 3 and 4 while $J \geq K$

Step 3: Set $LA[J+1] = LA[J]$

Step 4: Set $J = J - 1$

Step 5: Set $LA[K] = ITEM$

Step 6: Set $N = N + 1$

Step 7: Exit

C Program of Algorithm INSERT

```
int INSERT(int LA, int N, int K, int ITEM)
{
    int J;
    J=N-1; //Step 1
    while(J≥K) //Step 2
    {
        LA[J+1]=LA[J]; //Step 3
        J=J-1; //Step 4
    }
    LA[K]=ITEM; //Step 5
    N=N-1; //Step 6
    for(int I=0;I<N;I++)
        printf("%d, ",LA[I]);
    return N; //Step 7
}
```



Linear Array: Insertion and Deletion

.	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

← While executing

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

$N = 9$

For example, $K = 7$

$ITEM = 19$



Linear Array: Insertion and Deletion

.	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

While executing

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

$N = 9$
For example, $K = 7$
 $ITEM = 19$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

While executing

$N = 9$
For example, $K = 7$
 $ITEM = 19$
 $J = K = 7$
 $J \leq N - 1$ is true



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	19
J+1	20
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$ ← While executing

Step 5: Set $N = N - 1$

Step 6: Exit

$N = 9$
 $K = 7$
 $ITEM = 19$
 $J = 7$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	20
J+1	20
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$ ← While executing

Step 5: Set $N = N - 1$

Step 6: Exit

$N = 9$
 $K = 7$
 $ITEM = 19$
 $J = 7$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
J	20
J+1	20
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

While executing

$N = 9$
 $K = 7$
 $ITEM = 19$
 $J = 7 + 1 = 8$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
J	20
J+1	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

While executing

$N = 9$
 $K = 7$
 $ITEM = 19$
 $J = 7 + 1 = 8$
 $J \leq N - 1$
is True



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
J	20
J+1	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

While executing

$N = 9$
 $K = 7$
 $ITEM = 19$
 $J = 8$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
J	14
J+1	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

While executing

$N = 9$
 $K = 7$
 $ITEM = 19$
 $J = 8$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
J	14
J+1	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$ ← While executing

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

$N = 9$
 $K = 7$
 $ITEM = 19$
 $J = 8 + 1 = 9$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
J	14
J+1	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$ ← While executing

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

N=9
K=7
ITEM=19
J=9
 $J \leq N - 1$ is False



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
8	14
N	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$ ← While executing

Step 6: Exit

$N = 9 - 1 = 8$

$K = 7$

$ITEM = 19$

$J = 9$



Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
N	14
	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N - 1$

Step 4: Set $LA[J] = LA[J + 1]$

Step 5: Set $N = N - 1$

Step 6: Exit

While executing

$N = 10$

$K = 7$

$ITEM = 19$

$J = 9$

$N = 9 - 1 = 8$



CSE 203: Data Structure

Data
Structure

Linear Array: Insertion and Deletion

	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	20
N	14
	14

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set ITEM=LA[K]

Step 2: Print the value of ITEM

Step 3: Repeat for J=K to N-1

Step 4: Set LA[J]=LA[J+1]

Step 5: Set N=N-1

Step 6: Exit

While executing

N=10

K=7

ITEM=19

J=9

N=9-1=8



Linear Array: Insertion and Deletion

Algorithm DELETE:

Algorithm DELETE (LA, N, K, ITEM)

Here LA is a linear array with N elements and $1 \leq K \leq N$

The algorithm deletes Kth element from LA.

Step 1: Set $ITEM = LA[K]$

Step 2: Print the value of ITEM

Step 3: Repeat for $J = K$ to $N-1$

Step 4: Set $LA[J] = LA[J+1]$

Step 5: Set $N = N-1$

Step 6: Exit

C Program of Algorithm DELETE

```
int DELETE(int LA, int N, int K, int ITEM)
{
    int J;
    ITEM = LA[K]; //Step 1
    printf("%d", ITEM) //Step 2
    for(J = K; J < N-1; J++) //Step 3
        LA[J] = LA[J+1]; //Step 4
    N = N-1; //Step 5
    return N; //Step 6
}
```



CSE 203: Data Structure

Data
Structure

Linear Array: Sorting

.	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

Given an array LA. It contains N elements. The values of the elements are not stored in the array in any sorting order. Our task is to sort the elements in the array LA.



CSE 203: Data Structure

Data
Structure

Linear Array: Sorting

Bubble Sort Strategy:

.	LA
1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14



CSE 203: Data Structure

Data
Structure

Linear Array: Sorting

Bubble Sort Strategy:

. LA

1	10
2	12
3	11
4	15
5	21
6	22
7	19
8	20
N	14

After doing that

Checking all N elements, move the largest one to upward index position and finally put it here.

. LA

1	
2	
3	
4	
5	
6	
7	
8	
N	22

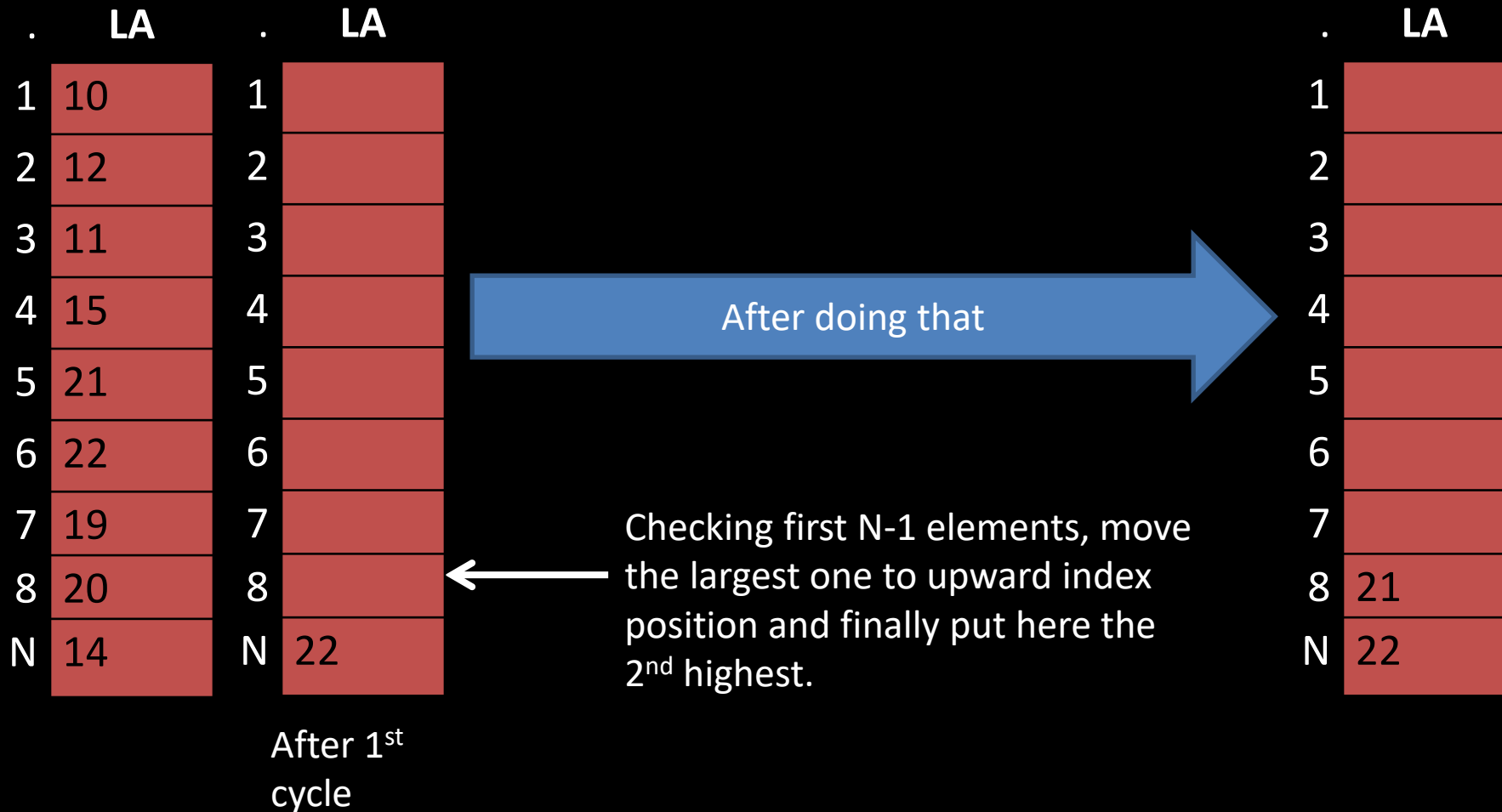


CSE 203: Data Structure

Data
Structure

Linear Array: Sorting

Bubble Sort Strategy:



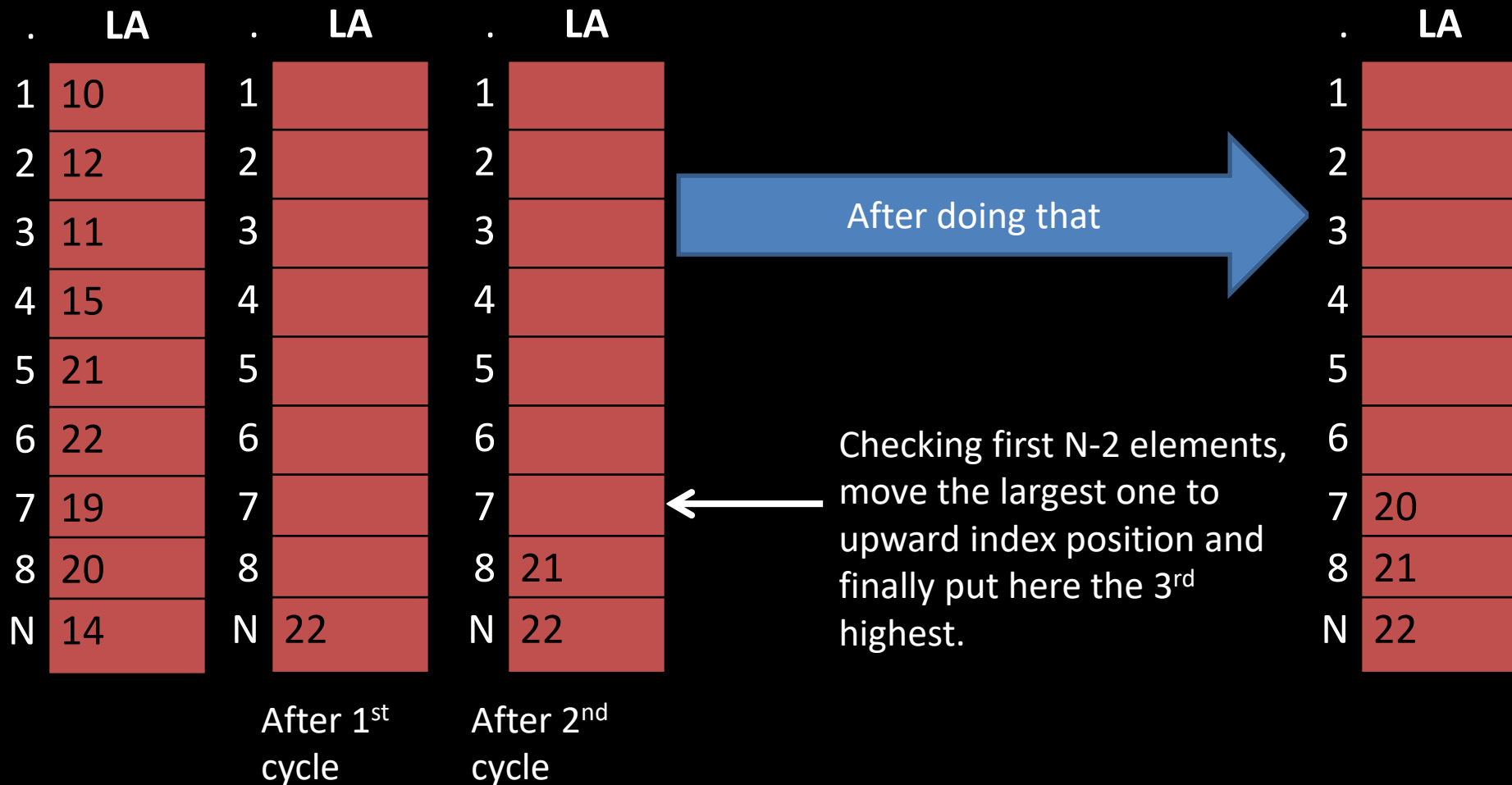


CSE 203: Data Structure

Data
Structure

Linear Array: Sorting

Bubble Sort Strategy:





Linear Array: Sorting

Suppose the following numbers are stored in an array A:

32, 51, 27, 85, 66, 23, 13, 57

We apply the bubble sort to the array A. We discuss each pass separately.
Pass 1. We have the following comparisons:

- (a) Compare A_1 and A_2 . Since $32 < 51$, the list is not altered.
- (b) Compare A_2 and A_3 . Since $51 > 27$, interchange 51 and 27 as follows:
32, (27), (51), 85, 66, 23, 13, 57
- (c) Compare A_3 and A_4 . Since $51 < 85$, the list is not altered.
- (d) Compare A_4 and A_5 . Since $85 > 66$, interchange 85 and 66 as follows:
32, 27, 51, (66), (85), 23, 13, 57
- (e) Compare A_5 and A_6 . Since $85 > 23$, interchange 85 and 23 as follows:
32, 27, 51, 66, (23), (85), 13, 57
- (f) Compare A_6 and A_7 . Since $85 > 13$, interchange 85 and 13 to yield:
32, 27, 51, 66, 23, (13), (85), 57
- (g) Compare A_7 and A_8 . Since $85 > 57$, interchange 85 and 51 to yield:
32, 27, 51, 66, 23, 13, (57), (85)



CSE 203: Data Structure

Data
Structure

Linear Array: Sorting

Pass 2. (27), (33), 51, 66, 23, 13, 57, 85

27, 33, 51, (23), (66), 13, 57, 85

27, 33, 51, 23, (13), (66), 57, 85

27, 33, 51, 23, 13, (57), (66), 85

Pass 3. 27, 33, (23), (51), 13, 57, 66, 85

27, 33, 23, (13), (51), 57, 66, 85

Pass 4. 27, (23), (33), 13, 51, 57, 66, 85

27, 23, (13), (33), 51, 57, 66, 85

Pass 5. (23), (27), 13, 33, 51, 57, 66, 85

23, (13), (27), 33, 51, 57, 66, 85

Pass 6. (13), (23), 27, 33, 51, 57, 66, 85



Linear Array: Sorting

Algorithm 4.4:

(Bubble Sort) BUBBLE(DATA, N)

Here DATA is an array with N elements. This algorithm sorts the elements in DATA.

1. Repeat Steps 2 and 3 for $K = 1$ to $N - 1$.
2. Set $PTR := 1$. [Initializes pass pointer PTR.]
3. Repeat while $PTR \leq N - K$: [Executes pass.]
 - (a) If $DATA[PTR] > DATA[PTR + 1]$, then:
 Interchange $DATA[PTR]$ and $DATA[PTR + 1]$.
 [End of If structure.]
 - (b) Set $PTR := PTR + 1$.
 [End of inner loop.]
- [End of Step 1 outer loop.]
4. Exit.



CSE 203: Data Structure

Data
Structure

Linear Array: Searching

Searching refers to the operation of finding an ITEM in a location LOC in an array DATA.

Output:

Successful

Unsuccessful

The algorithm that one chooses depends on the way the information in the DATA is organized.

Example:

Linear search

Binary search



Linear Array: Searching

Algorithm 4.5: LINEAR(DATA, N, ITEM, LOC)

Here DATA is a linear array with N elements, and ITEM is a given item of information. This algorithm finds the location LOC of ITEM in DATA, or sets $LOC := 0$ if the search is unsuccessful.

1. [Insert ITEM at the end of DATA.] Set $DATA[N + 1] := ITEM$.
2. [Initialize counter.] Set $LOC := 1$.
3. [Search for ITEM.]
Repeat while $DATA[LOC] \neq ITEM$:
Set $LOC := LOC + 1$.
[End of loop.]
4. [Successful?] If $LOC = N + 1$, then: Set $LOC := 0$.
5. Exit.

Complexity

$$O(n)=n$$



Linear Array: Searching

Algorithm 4.6: (Binary Search) BINARY(DATA, LB, UB, ITEM, LOC)
Here DATA is a sorted array with lower bound LB and upper bound UB, and ITEM is a given item of information. The variables BEG, END and MID denote, respectively, the beginning, end and middle locations of a segment of elements of DATA. This algorithm finds the location LOC of ITEM in DATA or sets LOC = NULL.

1. [Initialize segment variables.]
Set BEG := LB, END := UB and MID = INT((BEG + END)/2).
2. Repeat Steps 3 and 4 while BEG ≤ END and DATA[MID] ≠ ITEM.
3. If ITEM < DATA[MID], then:
Set END := MID - 1.
Else:
Set BEG := MID + 1.
[End of If structure.]
4. Set MID := INT((BEG + END)/2).
[End of Step 2 loop.]
5. If DATA[MID] = ITEM, then:
Set LOC := MID.
Else:
Set LOC := NULL.
[End of If structure.]
6. Exit.

Complexity

$$f(n) = \lfloor \log_2 n \rfloor + 1$$



$$O(n) = \lfloor \log_2 n \rfloor$$



CSE 203: Data Structure

Data
Structure

Linear Array: Searching **MULTIDIMENSIONAL ARRAYS**

Two-dimensional array A of size $m \times n$.

Each element of the array is specified by $A[J, K]$ where

$$1 \leq J \leq m, \quad 1 \leq K \leq n$$

		Columns			
		1	2	3	4
Rows	1	$A[1, 1]$	$A[1, 2]$	$A[1, 3]$	$A[1, 4]$
	2	$A[2, 1]$	$A[2, 2]$	$A[2, 3]$	$A[2, 4]$
	3	$A[3, 1]$	$A[3, 2]$	$A[3, 3]$	$A[3, 4]$

Student	Test 1	Test 2	Test 3	Test 4
1	84	73	88	81
2	95	100	88	96
3	72	66	77	72
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
25	78	82	70	85

Fig. 4.9 Array *SCORE*

`SCORE[2, 1], SCORE[2, 2], SCORE[2,3], SCORE[2, 4]`



CSE 203: Data Structure

Data
Structure

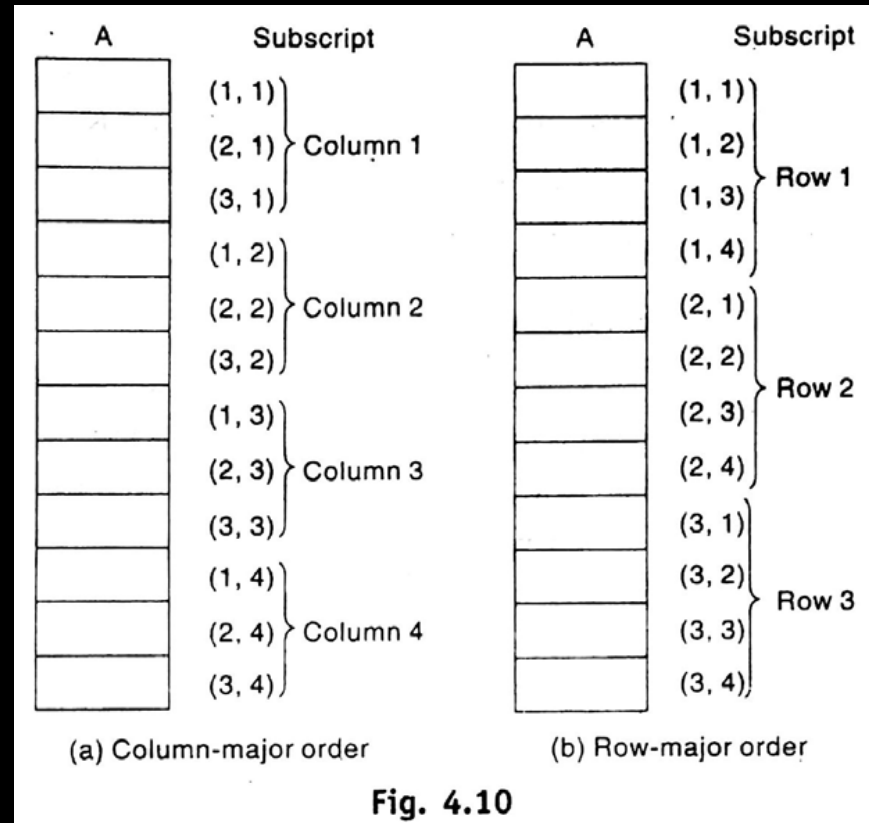
Linear Array: Searching **MULTIDIMENSIONAL ARRAYS**

Two-dimensional array A of size $m \times n$.

Each element of the array is specified by $A[J, K]$ where

$$1 \leq J \leq m, \quad 1 \leq K \leq n$$

Memory Representation:





CSE 203: Data Structure

Data
Structure

Linear Array: Searching **MULTIDIMENSIONAL ARRAYS**

Two-dimensional array A of size $m \times n$.

Each element of the array is specified by $A[J, K]$ where

$$1 \leq J \leq m, \quad 1 \leq K \leq n$$

$$B[K_1, K_2, \dots, K_N]$$